

UTEA Cover Letter: Xiaoying (Camilla) Xue

My research interest lies in pushing the boundaries of computer systems to improve performance across various domains. This interest is grounded in hands-on systems research where I analyzed low-level bottlenecks in database communication and storage engines, and implemented target optimizations with measurable performance gains. I am looking to extend this experience to broader performance-oriented problems, including CUDA kernel optimization for GPU computation, building user-facing applications that make complex data accessible, and advancing query optimization techniques for heterogeneous data types.

To ground my interest in low-level computer systems, I joined a research project supervised by Professor Qizhen Zhang and worked directly with the PostgreSQL codebase to analyze and improve database communication efficiency. My work involves tracing client-server interaction pathways and categorizing internal functions to investigate how system-level bottlenecks propagate through database layers. Building on this system-focused research experience, I participated in the CSC494 research group supervised by Professor Niv Dayan. In this project, I conducted comprehensive benchmarking of RocksDB, comparing binary-search and hash-based indexing approaches to quantify performance differences and identify read and write throughput bottlenecks. To improve query performance, I implemented a minimal perfect hash index integrated into the SST data-block write and read pipelines, optimizing index lookups and enhancing read throughput. This design achieved a consistent performance improvement of approximately 5% over existing indexing methods.

Drawing from the above experience, I would like to both broaden and apply my existing experience into a new setting of high performance computing. Specifically in the *High-Performance GPU Kernel Design for AI workloads* project led by Professor Maryam Mehri Dehnavi. While the prior research projects do not involve GPUs directly, the core processes of identifying bottlenecks, developing improvement strategies, implementing solutions, and benchmarking performance are highly transferable and directly applicable. To deepen my knowledge of GPU architecture and CUDA kernel, I participated in the *GPU Mode* reading group to build practical insight into GPU architecture. Motivated by discussions on profiling and hardware-aware scheduling, I familiarized myself with LLM computation demands and wrote related CUDA kernels for these workloads. For example, I implemented casual self-attention with both one- and two-dimensional thread block configurations to experiment on their performance.

I am equally driven to apply my skills to build impactful frontend tools that translate complex data into accessible insights for end users. This interest directly connects with the project led by Professor Alex Mariakakis, and I would like to contribute to solutions that advance healthcare research through improved data accessibility by combining my technical background with a focus on user-centered design. Given my prior experience working with database systems, I am interested in further exploring research at the intersection of databases and machine learning through Professor Nick Koudas's projects, which focus on optimizing LLM performance via strategic query execution plans and investigating the factors that influence LLM ranking behavior. This aligns with my systems knowledge and performance-driven mindset, allowing me to contribute to pioneering methods that enhance the efficiency of processing heterogeneous data types and develop systematic metrics to analyze ranking preferences in LLM.

Thank you for taking the time and considering my application.